KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Electrical and Electronic Engineering
Spring – 2024 Final Examination
Program: B.Sc. in EEE (Batch – 16^{th}) 1^{st} Year 2^{nd} Semester
Course Title: Physics - II
Course Code: PHY 0533 1201

PART-A: MCQ

Time: 10 Minutes **Total Marks: 10**

Student's ID: | | | | | | | | | | | | | | | | |

Date: | | | | | | | | | **Invigilator's Signature:** |

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

CLO1	Define the basic terms related to thermodynamics, modern physics and optics.	4

Cognitive Knowledge

1. The velocity of the light depends on the _____________ of the medium.
   A) number of charges
   B) resistivity
   C) rigidity modulus
   D) boiling point
   E) permeability

Cognitive Understanding

2. To form the bright fringe, what is the condition of the path difference between the two waves? [Where, Δl – path difference, λ – wavelength & n – integer number]
   A) $\Delta l = n(\lambda/2)$
   B) $\Delta l = n\lambda$
   C) $\Delta l = 2n\lambda$
   D) $\Delta l = (2n+1)\lambda$
   E) $\Delta l = (2n+1)\lambda/2$

Cognitive Knowledge

3. Which one is the alpha particle?
   A) ${}_{2}^{4}He$
   B) ${}_{0}^{1}n$
   C) ${}_{1}^{3}H$
   D) ${}_{1}^{2}H$
   E) ${}_{1}^{1}H$

Cognitive Understanding

4. Which is essential for interference?
   A) Two different color lights.
   B) The movement of the two waves should be opposite.
   C) The light sources should be far apart.
   D) The light sources should be coherent.
   E) The phase difference should have to be $\frac{\pi}{2}$.

CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	3

Cognitive Understanding

5. What is the mirror nucleus of ${}_{4}^{7}Be$?
   A) ${}_{2}^{4}He$
   B) ${}_{3}^{7}Li$
   C) ${}_{4}^{8}Be$
   D) ${}_{3}^{8}Li$
   E) ${}_{8}^{14}O$

Cognitive Understanding

6. Which nucleus have the maximum binding energy per nucleon?

A) Al
B) Xe
C) Fe
D) U
E) Au

Cognitive Knowledge

7. A train moving with a non-uniform speed is an example of …………………

A) 1st Postulate of special theory of relativity.
B) 2nd Postulate of special theory of relativity.
C) inertial frame of references.
D) non-inertial frame of references.
E) Lorentz transformation.

CLO3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and optics.	2

Cognitive Applying

8. What is the radius of the ${}^{27}_{13}\mathrm{Al}$ nucleus?

A) 2.6fm
B) 3.6fm
C) 4.6fm
D) 5.6fm
E) 6.6fm

Cognitive Applying

9. Half-life of a radioactive substance is 10 days. How long will it take to decrease to 25% of it?

A) 5 days
B) 10 days
C) 15 days
D) 20 days
E) 25 days

CLO4	Demonstrate analytical proficiency to establish the physical and mathematical logic behind natural phenomena observed in everyday life and propose various project ideas for technological advancement.	1

Cognitive Analysis

10. When the velocity v of the body is negligibly small as compared to light velocity c, what will be relation between stationary length, L_0 and moving length, L?

A) $L < L_0$
B) $L > L_0$
C) $L < L_0 < 0$
D) $L > L_0 > 0$
E) $L = L_0$

Signature of Examiner

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PART-B: SAQ

Time: 1 Hour and 50 Minutes **Total Marks: 30**

[Instruction: Marks for the questions are indicated against each of them]

CLO1	Define the basic terms related to thermodynamics, modern physics and optics.	5

[Answer any 1 (one) out of 2 (two) sets of questions]

1. a) Illustrate a binding energy per nucleon curve. 2
   b) Define: 1 amu, photo electron & inertial frame. 3

2. a) State Huygens principle. 2
   b) Specify diffraction of light, Polarization of light and wave front. 3

CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	10

[Answer any 2 (two) out of 3 (three) sets of questions]

3. a) Formulate speed of electromagnetic wave. 2
   b) Explain how constructive and distractive interference of light occur. 3

4. a) Compare two photons of Compton's scattering. 2
   b) Write a short note on "light possesses waves as well as particles properties." 3

5. a) Distinguish fission and fusion reaction. 2
   b) Make a logic to explain the velocity of any object cannot be more than the velocity of light. 3

CLO3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and optics.	10

[Answer any 2 (two) out of 3 (three) sets of questions]

6. a) Interpret the nuclear chain reaction and mention the controlling mechanism of it. 2
   b) Calculate the value of the energy of the following nuclear reaction: 3

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow (^{236}_{92}\mathrm{U})^{*} \rightarrow ^{141}_{56}\mathrm{Ba} + ^{92}_{36}\mathrm{Kr} + 3^{1}_{0}\mathrm{n} + \mathrm{Energy}$$

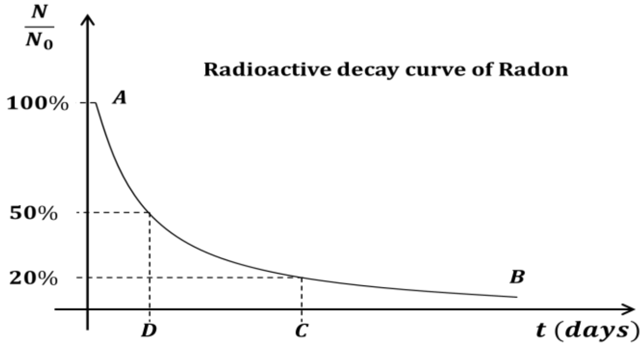
Where, $M(^{235}_{92}U) = 235.0439amu$; $M(^{141}_{56}Ba) = 140.9139amu$;
$M(^{92}_{36}Kr) = 91.8973amu$; $M(^{1}_{0}n) = 1.0087amu$

7. a) How does classical physics fail to explain the photoelectric effect? 2
   b) Find the maximum velocity of photo-electrons emitted by radiation of frequency 3×10^{15} Hz from a photoelectric surface having a work function 4.0 eV. 3

8. a) Established the relation between the total energy and momentum. 1
   b) A space shuttle crosses the football ground of BUET along of its length at speed of 0.6c. The football ground is 100m long and 50m wide. What is the length and width of the football ground according to the measurement by a foreign astronaut in the space shuttle? 4

CLO4	Demonstrate analytical proficiency to establish the physical and mathematical logic behind natural phenomena observed in everyday life and propose various project ideas for technological advancement.	5

9. See the stem given below and answer the following questions:



a) What is the value of point D? 1

b) Find the value of point C. 2

c) Is it possible to touch the point B with X axis? – Justify your answer. 2